

sample would not be used, but the raw mean would be adjusted to the mean implied by the CAPM (see chapter 6), a multifactor model, or some value computed from fundamental analysis. Covariances can also be shrunk back to a prior where each stock in an industry, say, has the same volatility and correlation—which is reasonable if we view each stock in a given industry as similar to the others.²⁵

No statistical method, however, can help you if your data are lousy.

Don't Just Use Historical Data

Investors must use past data to estimate inputs for optimization problems. But many investors simply take historical averages on short, rolling samples. This is the worst thing you can do.

In drawing all of the mean-variance frontiers for the G5, or various subsets of countries, I used historical data. I plead guilty. I did, however, use a fairly long sample, from January 1970 to December 2011. Nevertheless, even this approximately forty-year sample is relatively short. You should view the figures in this chapter as what has transpired over the last forty years and not as pictures of what will happen in the future. As the investment companies like to say in small print, past performance is no guarantee of future returns. The inputs required for mean-variance investing—expected returns, volatilities, and correlations—are statements about what we think will happen in the future.

Using short data samples to produce estimates for mean-variance inputs is very dangerous. It leads to pro-cyclicality. When past returns have been high, current prices are high. But current prices are high because future returns tend to be low. While predictability in general is very weak, chapter 8 provides evidence that there is some. Thus, using a past data sample to estimate a mean produces a high estimate right when future returns are likely to be low. These problems are compounded when more recent data are weighted more heavily, which occurs in techniques like exponential smoothing.

An investor using a sample where returns are stable, like the mid-2000s right before the financial crisis, would produce volatility estimates that are low. But these times of low volatilities (and high prices) are actually periods when risk is high. Sir Andrew Crockett of the Bank of England says (with my italics):²⁶ “The received wisdom is that risk increases in the recessions and falls in booms. In contrast, it may be more helpful to think of risk as *increasing* during upswings, as financial imbalances build up, and *materializing* in recessions.” The low estimates

²⁵ See Ledoit and Wolf (2003) and Wang (2005), among others. Strictly speaking, the mean-variance solution involves an inverse of a covariance matrix, so we should shrink the inverse covariance rather than the covariance. This is done by Kourtis, Dotsis, and Markellos (2009). Tu and Zhou (2011) show shrinkage methods can be used to combine naïve and sophisticated diversification strategies in the presence of estimation risk.

²⁶ “Marrying the Micro- and Macro-Prudential Dimensions of Financial Stability,” speech at the Eleventh International Conference of Banking Supervisors, Sept. 20–21, 2000, <http://www.bis.org/speeches/sp000921.htm>.